Health Impacts of Air Pollution in Indoor Tunnels: A Focus on Carbon Monoxide and Particulate Matter

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Abstract

Air pollution poses a significant health risk, particularly in enclosed spaces like road tunnels where vehicle emissions accumulate. This study investigates whether pollutant levels in Boston tunnels meet health standards, evaluates the efficiency of the ventilation system, and examines the impact of different vehicle types on air quality. Carbon monoxide (CO) and particulate matter (PM), two major pollutants emitted by vehicles, were measured in selected tunnels. Results indicate that pollutant levels generally remain within healthy thresholds, suggesting the ventilation system is effective under normal conditions. However, during prolonged traffic and in the presence of high-emission vehicles, such as trucks, pollutant levels exceeded safe limits. These findings highlight the need for improved ventilation systems to ensure air quality remains safe under diverse traffic conditions, highlighting the importance of continuous monitoring and system improvements to mitigate health risks in indoor tunnels.

Introduction

Air pollution remains one of the most pressing global health challenges, contributing to approximately 7 million deaths annually and accounting for 1 in 9 deaths worldwide, making it the largest environmental health risk, according to the World Health Organization (WHO Report, 2017). The economic impact is equally critical, with annual losses of $8.1 trillion, representing nearly 5% of global GDP (World Bank, 2022). Beyond health and economic concerns, air pollution exacerbates climate change, harms biodiversity, and diminishes crop yields. Within urban areas, road tunnels are particularly concerning as they pose a higher risk associated with air pollution.

Road tunnels differ significantly from open roads or urban streets due to the confined nature of these environments. Vehicle emissions, responsible for up to 75% of urban air pollution (EPA, 2015) and 14% of global greenhouse gas emissions (IPCC, 2019), become highly concentrated in enclosed spaces. Pollutant levels in road tunnels can reach 2-4 times higher concentrations than on open roads due to poor ventilation systems, leading to the accumulation of harmful pollutants such as carbon monoxide (CO), particulate matter ($PM_{2.5}$), and nitrogen oxides (NO_x) (Zhou et al., 2014). Additionally, the risks of tunnel fires and emergencies further underscore the need for efficient air quality management systems in these settings (Gehandler et al., 2015).

Despite increasing awareness of air pollution, significant gaps remain in understanding its dynamics within specific tunnel systems, including those in Boston. With the city's growing commuter population and increased vehicular traffic, pollutant levels are likely to have changed. However, studies focusing on the ventilation systems in Boston's tunnels and the impact of different vehicle types on pollution levels are limited. Addressing this knowledge gap is essential, given the critical role tunnels play in urban transportation and the potential health risks posed by poor air quality in enclosed spaces.

This study aims to measure pollutant levels in Boston's tunnels, evaluate their compliance with health standards, and assess the efficiency of existing ventilation systems. By analyzing fluctuations in pollutant levels under varying traffic conditions and vehicle types, the research seeks to determine the capacity of these systems to manage emergencies and prolonged congestion. It is hypothesized that Boston's tunnels predominantly use longitudinal ventilation systems, which perform effectively under moderate and moving traffic conditions. However, under certain scenarios—such as heavy traffic in long tunnels or emissions from high-pollution

vehicles like trucks—these systems may struggle to maintain pollutant concentrations within healthy thresholds.

This research has direct implications for public health, as it provides insights into reducing the exposure of commuters to harmful pollutants. Moreover, findings can inform transportation and urban planning policies, aiding cities like Boston and others with similar infrastructure in enhancing tunnel safety and air quality management. By addressing the intersection of health, environmental, and urban challenges, this study contributes to a safer and more sustainable urban environment.

Methods

Study Location and Sampling Design

Figure 1
Map of Boston Tunnels
Note. Adapted from "Ted Williams Tunnel," Wikipedia, 10 Nov. 2024

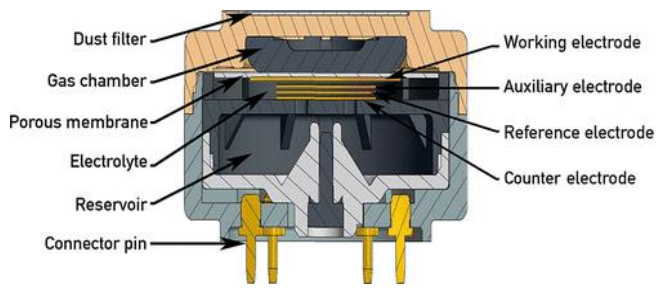
Data collection was conducted during a 45-minute trip through Boston's interconnected road tunnels, chosen for their enclosed structure and high traffic volume. These conditions provided a controlled environment to study pollutant behavior under typical urban traffic scenarios. Measurements included carbon monoxide (CO), nitrogen oxides (NO_x), ozone (O_3), particulate matter ($PM_{1.0}$ and $PM_{2.5}$), and carbon dioxide (CO_2). However, the analysis primarily focused on CO, $PM_{1.0}$, and $PM_{2.5}$, as these pollutants are key indicators of air quality and are directly linked to vehicle emissions.

The sampling route covered a series of tunnels representing diverse traffic conditions. Observational notes recorded external factors such as traffic density, vehicle types, and proximity to high-emission vehicles like trucks. These contextual observations were critical for interpreting pollutant levels and understanding the environmental factors influencing air quality.

Instrumentation and Pollutant Detection

Carbon Monoxide (CO) Detection

Figure 2
Design for standard amperometric gas sensors
Note. Adapted from ACS Sensors



A high-precision Alphasense electrochemical sensor was used to measure CO concentrations in real time. The sensor functions by oxidizing CO at a working electrode, generating an electric current proportional to the CO concentration. An auxiliary electrode monitored background noise, enhancing accuracy by subtracting unrelated currents. To maintain optimal reaction conditions, the sensor controlled the potential at the working electrode, ensuring precise and proportional measurements.

Particulate Matter (PM) Detection

Particulate matter ($PM_{1.0}$ and $PM_{2.5}$) concentrations were measured using the Senseair K30 laser scattering sensor. The sensor operates by emitting a laser beam into a sensing chamber, where airborne particles scatter the light. A photodetector captures the scattered light, and the intensity and angle of scattering are analyzed to calculate particle size distribution and concentration. The sensor's ability to provide real-time data on PM levels was essential for evaluating air quality and identifying pollution sources.

GPS Data Collection

An app called Phyphox was used alongside pollutant sensors to log the location and time of measurements. This data allowed for precise mapping of pollutant levels within the tunnel system, facilitating the identification of high-pollution zones and correlating air quality with specific traffic patterns and tunnel sections.

Data Integration

The integration of pollutant monitoring, GPS location data, and contextual observational notes ensured a robust and comprehensive dataset. Real-time pollutant measurements, combined with location and time data, enabled detailed analysis of how traffic conditions, vehicle types, and ventilation efficiency influenced air quality. This approach provided critical insights into the dynamics of air pollution in enclosed tunnel environments.

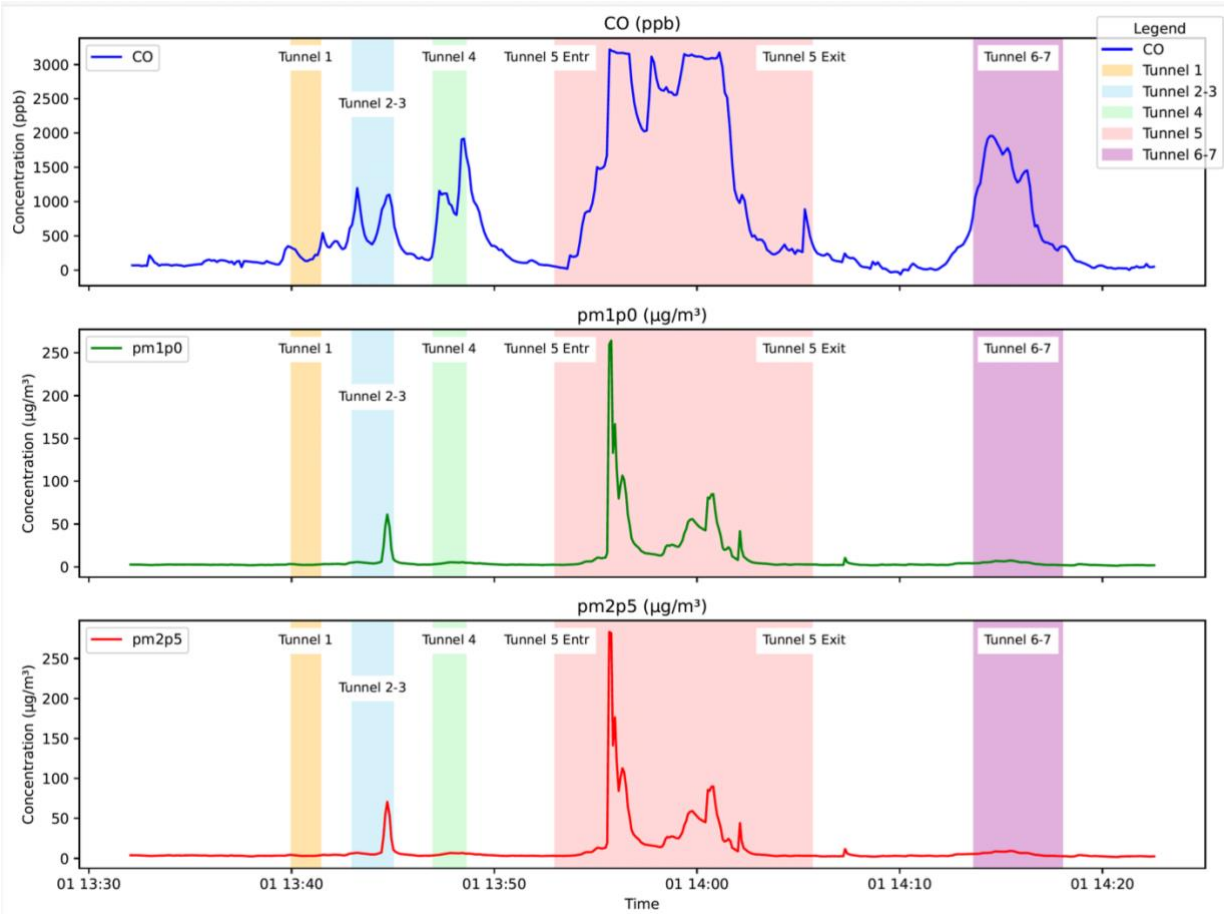
Results

The measurements of pollutant concentrations, including carbon monoxide (CO) and particulate matter ($PM_{1.0}$ and $PM_{2.5}$), collected during the study are presented in Figure 1. The times spent inside Boston's road tunnels are highlighted on the plots, allowing for a direct comparison of pollutant levels within and outside the enclosed spaces. Pollutant concentrations are displayed as absolute values, with CO reported in parts per billion (ppb) and PM in micrograms per cubic meter ($\mu g/m^3$). These values were exported directly from sensor readings obtained via ThingSpeak Cloud. Google Colab was used to plot the data.

Figure 3

Time dependence of concentrations of CO, PM1.0, and PM2.5

Note. Regions corresponding to times inside the tunnel are highlighted.



The time spent in tunnels was identified using GPS data collected during the field trip, combined with observational notes of tunnel entry and exit times. The recorded measurements span the entire 45-minute trip, with timestamps displayed in standard time format (e.g., 13:34 for the start and 14:25 for the end), providing clear temporal context for pollutant trends.

The data reveals several distinct trends in pollutant levels. For CO, concentrations are noticeably elevated at the entrances and exits of tunnels 1, 2-3, and 4, suggesting a buildup of emissions at these transition points. Tunnel 5 displays a sharp peak in CO levels. In tunnels 6-7, CO levels are initially high near the entrances but gradually decrease toward the exits.

Particulate matter concentrations ($PM_{1.0}$ and $PM_{2.5}$) remain near zero for most of the measurement period, indicating relatively low levels of particulate pollution in the tunnels overall. However, tunnel 5 stands out with a sharp increase in PM levels, coinciding with the peak observed in CO concentrations.

These results demonstrate that pollutant levels in Boston's road tunnels exhibit significant spatial and temporal variation, influenced by factors such as tunnel design, ventilation efficiency, and traffic conditions. The observations provide valuable insight into the distribution of air pollutants in enclosed urban environments and form a critical foundation for evaluating the performance of existing ventilation systems and identifying areas for improvement.

Discussion

Health Implications

The concentrations of carbon monoxide (CO) and particulate matter ($PM_{1.0}$ and $PM_{2.5}$) measured in Boston's tunnels were evaluated against the World Health Organization (WHO) thresholds to assess potential health risks. Table 1 outlines CO levels, their corresponding health effects, and comments on exposure. Our results show that CO concentrations in most parts of the tunnels remained under 3 ppm, which, according to Table 1, falls within the "healthy" region. This suggests no significant immediate risk of CO poisoning for the general population.

Table 1
Carbon Monoxide Levels Table
Note. Created from WHO data

Concentration Level	Effects of CO Poisoning	Comments
0-1 PPM	none, no symptoms	considered to be normal background levels
1-9 PPM	none - monitor air closely	maximum indoor air quality - short term
9-35 PPM	minimal effects possible over time	OSHA - 8 hour limit
35-100 PPM	flu-like symptoms, headache	mild exposure, but potentially dangerous
100-200 PPM	dizziness, drowsiness, vomiting	OSHA - 15 minute exposure limit
200-400 PPM	frontal headache, fatigue	medium exposure, but not safe to occupy
400-800 PPM	unconsciousness, brain damage	extreme exposure, life-threatening after 3 hrs.
800-1600 PPM	nausea, convulsions, death	very dangerous and hazardous - death in 1 hr.
12,000 PPM	death in 1 to 3 minutes	extremely hazardous environment

Similarly, Table 2 provides thresholds for particulate matter (PM) concentrations and their associated health impacts. It is important to note that these WHO thresholds are based on 24-hour exposure limits. While our data showed PM peaks of 275 $\mu g/m^3$ and 100 $\mu g/m^3$ at certain points in the tunnels, these spikes lasted only about a minute. When averaged over a 24-hour period, the PM concentrations are close to zero, well below hazardous levels.

Table 2
Particulate Matter Levels Table
Note. Created from WHO data

AQI Category	Index Values	Revised Breakpoints ($\mu g/m^3$, 24-hour average)
Good	0 - 50	0.0 - 12.0
Moderate	51 - 100	12.1 - 35.4
Unhealthy for Sensitive Groups	101 - 150	35.5 - 55.4
Unhealthy	151 - 200	55.5 - 150.4
Very Unhealthy	201 - 300	150.5 - 250.4
Hazardous	301 - 500	250.5 - 500

Overall, the comparison of measured pollutant concentrations with WHO thresholds indicates that air quality in Boston's tunnels generally meets health standards. However, the brief peaks in pollutant levels, particularly for PM, may require further investigation, especially for vulnerable populations or during prolonged exposure scenarios such as traffic congestion.

CO Levels Across Tunnel Sections

Table 3 summarizes average CO concentrations at the entrance, middle, and exit of each tunnel. A clear pattern emerges for tunnels 1, 2-3, and 4, where CO levels are significantly higher at entrances and exits compared to the middle. This pattern provides important insights into the ventilation systems used in these tunnels. Specifically, the data supports the hypothesis that most Boston tunnels utilize a longitudinal ventilation system, which tends to draw pollutants out through tunnel openings but is less effective at uniformly dispersing emissions within the tunnel. In contrast, PM concentrations remained close to zero for most of the tunnels, indicating that particulate matter is less affected by the ventilation system or the enclosed environment. This suggests that PM pollution in tunnels may be driven more by isolated, high-emission events rather than the overall traffic flow or ventilation inefficiencies.

Table 3

Average CO Levels for Each Tunnel and Their Respective Ratios

Note. CO levels are presented in parts per million (ppb), and ratios

	CO levels [ppb]			ratios	
	entance	middle	exit	entrance/middle	exit/middle
tunnel 1	400	100	500	4	5
tunnel 2-3	1100	400	1000	2.75	2.5
tunnel 4	1100	800	1800	1.4	2.25
tunnel 5	100	3000	800	0.033	0.27
tunnel 6-7	1000	1500	100	0.67	0.067

Peaks in Tunnel 5

As highlighted in the results and shown in Figure 1, unusual peaks in both CO and PM concentrations were observed upon entering Tunnel 5. Table 3 further confirms this, with CO concentrations reaching 3 ppm in the middle of the tunnel, significantly higher than other tunnels. Unlike the patterns in other tunnels, these peaks cannot be attributed to the ventilation system or ambient pollutant levels.

Field notes indicate that these peaks occurred when the vehicle carrying the sensors was stationary in traffic behind a truck. Trucks are known to emit significantly more CO and PM than passenger vehicles, and idling conditions increase the emissions. This observation underscores the disproportionate impact of heavy-duty vehicles on air quality within enclosed spaces and highlights the importance of managing emissions from such sources.

Improvements for Future Work

The data collected in this study was influenced by external factors, such as the truck emission event in Tunnel 5. To improve reliability and accuracy, future experiments should involve repeated measurements over multiple days and under varying conditions. Conducting measurements during peak traffic hours, such as rush hour, would also help assess whether pollutant exposure remains within healthy levels during worst-case scenarios. These additional data points would enhance the robustness of the conclusions and provide a clearer understanding of air quality dynamics in tunnels.

Vulnerable Groups

It is crucial to acknowledge that the WHO thresholds for CO presented in Table 1 are based on healthy adults and primarily account for cardiovascular risks. However, numerous studies have shown that CO exposure can also contribute to neurodevelopmental issues, making children and pregnant women particularly vulnerable (Cooper et al., 2012). Even brief exposure to elevated CO levels could have adverse effects on these groups, highlighting the need for stricter air quality monitoring and targeted mitigation efforts in environments such as tunnels.

Policy Recommendations

One potential improvement is transitioning from the current longitudinal ventilation system to a transverse ventilation system, which ensures uniform air quality throughout the tunnel and is more efficient in handling heavy traffic and long tunnels. Additionally, implementing stricter emission controls for heavy-duty vehicles, particularly during peak traffic hours, could significantly reduce pollutant levels. Policies encouraging the use of electric or low-emission vehicles in enclosed spaces like tunnels could further improve air quality and protect public health.

Conclusion

This study highlights the complex dynamics of air pollution in Boston's road tunnels, focusing on carbon monoxide (CO) and particulate matter ($PM_{1.0}$ and $PM_{2.5}$) as critical indicators of air quality. The results demonstrate that, under normal traffic conditions, pollutant levels generally comply with health standards, underscoring the effectiveness of the current ventilation systems. However, peaks in pollutant concentrations during heavy traffic or in the presence of high-emission vehicles, such as trucks, reveal the limitations of existing systems under certain

scenarios. These findings underscore the importance of continuous monitoring and targeted improvements in tunnel ventilation to address transient but potentially harmful spikes in pollution.

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